Jesse Hoppo

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SKILLS

- **Languages**: C/C++, Python, R, Java, SQL
- **Frameworks & Libraries**: React, Next.js, Django/Wagtail, Prisma, Auth.js
- **DevOps & Infra**: Git, Docker, Podman, Terraform, pyinfra, Azure DevOps, CI/CD, ELK, Filebeat, JFrog Artifactory
- **Systems & Platforms**: Linux, RHEL, Ubuntu, Networking, Distributed Systems, Concurrency, systemd
- **Tools**: Jira, Confluence, Wireshark, GitHub, Neon PostgreSQL, Vercel

EXPERIENCE

Software Engineer Intern | **Australian Energy Market Operator** | **Dec 2025 – Present**

- Designed and deployed ELK logging infrastructure on RHEL 9 servers using Pyinfra, Podman, Filebeat, Journald, and Logstash, improving deployment consistency and observability across production and pre-production forecasting services.
- Owned an end-to-end proof of concept for an ML model registry using JFrog Artifactory, evaluating model storage, versioning, RBAC, authentication, access control and model-scanning with Jfrog Xray.
- Built automated validation tooling using Git pre-commit hooks and R directory hashing, reducing manual verification effort and helping prevent invalid merges across SQL and Oracle data.
- Improved Docker-based Azure DevOps CI/CD pipelines for various codebases, reducing pipeline failure rates by 10% and improving release reliability.
- Refactored SMTP scripts into a reusable internal library for the operational team, streamlining BAU workflows and saving approx. 5+ hours a week.
- Authored internal Confluence documentation covering Git, proxy networking, Active Directory fixes, UML, and related engineering processes to improve onboarding and team knowledge.

Software Developer, POC | **Capstone Project / CEIH** | **Feb 2026 – July 2026**

- Led agile delivery as Scrum Master, coordinating sprint planning, stand-ups, backlog prioritization, task tracking, and retrospectives using Jira
- Evaluated and justified the adoption of Wagtail CMS, a Python/Django-based platform, against requirements for scalability, security, accessibility, maintainability, and ease of use.
- Designed portal workflows, content structures, and data models to support clinician-facing content management, user interactions, and future system extensibility.
- Developed and configured Wagtail CMS components, including content models, page structures, admin functionality, and database interactions to meet the core functional requirements gathered through stakeholder analysis.
- Created a project risk management framework to identify, monitor, and mitigate technical risks

Open-Source Contribution | **PyInfra**

- Contributed to PyInfra, an infrastructure automation tool used to provision and manage Linux systems through Python-based scripts.
- Developed a feature to allow hidden values to be passed through PyInfra environment variables, allowing secrets to be properly masked during execution and bash logs.
- Collaborated and communicated with project maintainers and community members through GitHub issues and Pull Request workflows, using local and dev-container testing to validate behavior before proposing upstream changes.

PROJECTS

Personal Website: https://jesse-hoppo.netlify.app/ (For any additional information or projects)

Inflow – Invoice follow-up (https://inflow-demo.vercel.app/) | Next.js, Tailwind CSS, Neon PostgreSQL, Auth.js

- Built and deployed a full-stack invoice management SaaS for managing clients, invoices, overdue payments, follow-up workflows, and dashboard analytics.
- Configured separate development and production environments using Neon PostgreSQL, supporting safer testing and schema changes.
- Designed the application structure around maintainable full-stack patterns, including typed data models, authentication, protected routes, and reusable UI components.

Dead Man's Draw – C++ card game (https://github.com/Jesse-11/DeadMansDraw) | C++, OOP, STL, Memory Management

- Built a playable C++ card game based on Dead Man's Draw, implementing deck management, card drawing, scoring, and replayable game rounds.
- Designed modular object-oriented components for cards, players, game state, turn flow, and rule handling using classes, encapsulation, control flow, and STL containers.
- Implemented input validation, game-loop control, shuffled card generation, and score calculation to create a playable console executable.

EDUCATION

University of Adelaide | **Adelaide, SA**

Bachelor of Software Engineering (Honours), Major in Smart Technologies | Feb 2023 – Nov 2026

- Computer Science Club (CSC) – collaborated in social events and contributed to open-source initiatives.
- Engineers Australia – accredited degree; participated in EA events.
- Coursework – Data Structures, Operating Systems, Embedded Systems, Networks, Web & Database Systems, Artificial Intelligence, Systems Engineering.